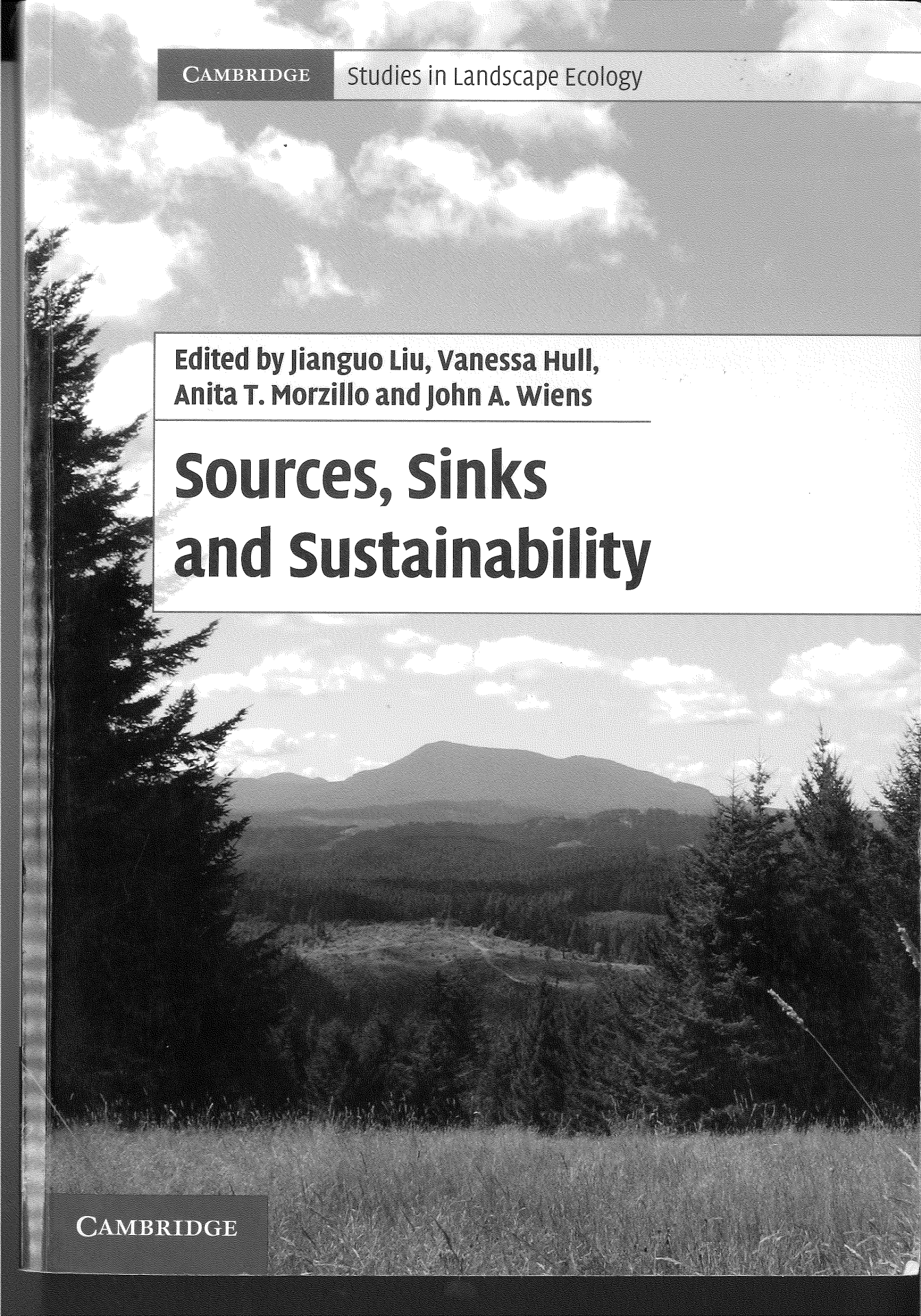




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Studies in Landscape Ecology

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Sources, Sinks and Sustainability

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When sinks rescue sources in dynamic environments

Summary

Many species of conservation concern occur in spatially heterogeneous landscapes composed of different patches that function as population sources and population sinks. Temporal variation in habitat quality, due to a cycle of habitat disturbance and subsequent recovery, can create relatively underappreciated source–sink dynamics. A cycle of disturbance and recovery can cause a given patch to alternate between functioning as a population source and a population sink. During “good” years, this patch can conceivably sustain another nearby patch that is always sink habitat. However, after a disturbance, the putative source patch may then depend upon individuals from that sink for recolonization. Thus, the metapopulation can depend upon the presence of sink populations for long-term persistence, provided that the sink is relatively unaffected by disturbance. We developed a simple, two-patch model of source–sink dynamics in order to explore the sensitivity of long-term metapopulation persistence to the temporal scale of habitat disturbance, recovery in a putative source patch, and the rate of population decline in the sink. We found that management directed at decelerating the rate of population decline in the sink can have a much greater affect on metapopulation persistence than management targeted at increasing the rate of habitat recovery in the source. This result is magnified as disturbance frequency increases. It is hoped that sinks crucial to metapopulation survival are given appropriate conservation status.

Introduction

Modeling population dynamics has played a prominent role in the history of ecology (Kingsland 1995). Many early models assume, for convenience only, that space is homogeneous. This began to change as the theory of island biogeography and metapopulation concepts grew. By the mid-1980s, many ecologists were concerned with spatial variation in habitat at the landscape scale. Thus, ecologists quickly embraced source-sink dynamics, which is predicated on the fact that landscapes are composed of heterogeneous habitats and that demographic parameters are habitat-specific. The original formulations of source-sink dynamics (Holt 1985; Pulliam 1988) make a clear and convincing case for the consequences of spatial variation of habitat quality on population dynamics without invoking temporal variability. These models illustrate that spatial differences in demographic rates, when coupled with immigration and emigration, lead to density-dependent limits to population growth. This is important. Source-sink models essentially use space rather than time to explain how populations self-regulate. But the exclusion of temporal variation in the early models is somewhat unrealistic because conditions clearly vary across seasonal, yearly, and decades-long intervals of disturbance and succession.

Source-sink models have been developed that incorporate temporal variation of the underlying spatial differences in demographic parameters among habitats. Holt (1997; Chapter 2, this volume) showed that dispersal into sink habitats is an evolutionarily stable strategy if source fitness periodically falls below sink fitness, and descendants of dispersers can move back to the source. Holt (1997) further demonstrated that utilization of sink habitat increases with increasing sink fitness and deviation from carrying capacity in the source. Gonzalez and Holt (2002) and Holt *et al.* (2003) demonstrated that autocorrelation of temporal variation in the growth rate of a sink population increases the average sink population size, a phenomenon known as the "inflationary effect." This finding was extended to an ensemble of connected sink patches, where temporal autocorrelation of the growth rates within patches was shown to permit metapopulation persistence (Roy *et al.* 2005). Foppen *et al.* (2001) used simulations to show that the resilience of a source after a catastrophe was enhanced by the presence of a sink, and empirical work on temporal variation in source-sink beetle dynamics has demonstrated that floods can transform sources into transient sinks (Johnson 2004). Gyllenberg *et al.* (1996) and Morris (Chapter 3, this volume) have also shown that a sink can rescue a source from extinction. In this chapter, we extend this idea to an analysis of management options for scenarios with different disturbance frequencies, rates of source habitat recovery, and population decline in the sink.

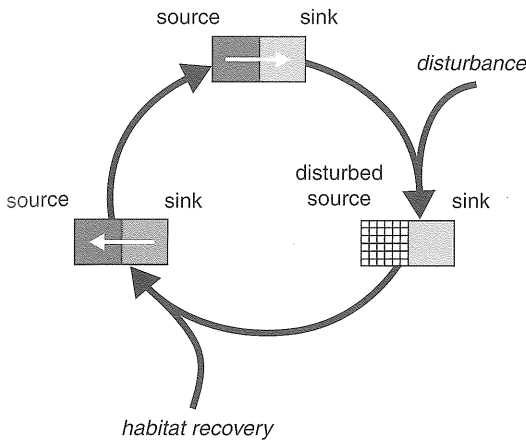


FIGURE 7.1.

Source-sink dynamics where disturbances affect one of two populations. Before disturbances occur, a typical source-sink relationship exists between two populations (topmost portion of figure). Movement of individuals from the source to the sink (white arrow) maintains the sink population. A disturbance degrades the source, putting the system in a critical period where both populations are at risk. Once the source habitat has recovered it is repopulated by individuals dispersing from the sink, eventually restoring the system to the original source-sink scenario.

In this chapter, we explore the effects of a particular form of temporal variability in source-sink dynamics. Specifically, we are concerned with the effects of recurrent disturbances that devastate a source but leave an associated sink relatively undisturbed. We model a simple system of two populations linked by dispersal. In the absence of disturbances, one population occurs in a habitat that is a permanent source while the other population occurs in a sink. Disturbances reduce population size in the source and destroy source habitat, thereby placing the source population in jeopardy until the habitat is sufficiently recovered. This process of habitat destruction and renewal is depicted in Figure 7.1. Note that the system enters a critical period after disturbances, wherein metapopulation persistence crucially depends on the recovery and recolonization of source habitat before the sink population goes extinct. Thus habitat availability, not population size, limits population growth and recovery. Here we explore the interactions among three rates that are critical to long-term metapopulation viability in this system:

1. the rate of habitat recovery in the source
2. the rate of population decline in the sink
3. the frequency of disturbances affecting the source.

The interactions between these factors are important because they have direct relevance to the identification of best management actions for the following dilemma: Should conservation and recovery efforts be devoted to increasing

the rate of post-disturbance habitat recovery in the source or decreasing the rate of population decline in the sink? Which of these two rates has the greatest impact on population viability? We can add another layer of complexity by asking: How does the relative influence of these rates on population viability change as the disturbance frequency changes? Our objective in this chapter is to provide an answer to these questions.

Example systems

In order of increasing generality, we provide three examples of systems that may exhibit the source-sink dynamics described in this chapter. A common feature of these systems is that all sources are destroyed at the same time and sinks are relatively unharmed. A system with just one source is also relevant (indeed, we model a system with one source and one sink). If at least one source was unaffected by disturbance, then recovery of the metapopulation would be largely driven by dispersal out of that source. Thus, a key component of the systems we describe is temporal synchrony of source destruction. The systems we describe are likely to occur where characteristics conferring enhanced productivity upon a habitat are correlated with the occurrence of disturbances.

Alabama beach mouse and hurricanes

Our formulation of this particular source-sink dynamic is motivated by our involvement with the Alabama beach mouse (*Peromyscus polionotus ammobates*). This subspecies of the oldfield mouse became a federally listed endangered species in 1985. Its entire historic range is limited to an approximately 48 km strip of Alabama's coveted beachfront real estate. Less than 40% of the 2,830 hectares (ha) of pre-development habitat remains (USFWS 2005). Habitat loss and fragmentation certainly threaten the Alabama beach mouse's existence, and the severity of this problem is dramatically magnified when hurricanes strike.

Beach mice make burrows and forage for seeds and insects in sandy dunes lightly vegetated by sea oats (*Uniola paniculata*), blue stem (*Andropogon maritimus*), and beach grass (*Panicum amarum*). The dunes gradually increase in height (and age) until a well-defined dune apex is reached at roughly 250 meters (m) from the open ocean. On the other side of this dune ridge is a very distinct habitat type: soils are more compact and the vegetation is dominated by sand live oak (*Quercus geminata*), saw palmetto (*Serenoa repens*), and rosemary (*Ceratiola ericoides*). This area is referred to as "scrub." Beach mice were once thought to inhabit dunes exclusively (Howell 1909, 1921; Ivey 1949), but it is now known that beach mice also occur at low densities in scrub adjacent to the dunes (Sneckenberger 2001).

In addition to differences in substrate density, vegetation type, and beach mouse density between the dunes and scrub, the two habitat types also differ in susceptibility to hurricane-induced inundation. On September 16, 2004, Hurricane Ivan was at the boundary of a category 3–4 storm when it made landfall directly over the Alabama beach mouse's entire range. The storm created a 3.0–4.5 m surge ("surge" is quite literally a moving hill of water created by hurricane-strength wind) that obliterated the dunes but left the scrub relatively intact. At the time of writing this, beach mice are largely absent from dunes that do not have scrub behind them, suggesting that the scrub provides refugia from storms and acts as a source of colonists for the recovering dune habitat. Indeed, 47% of the mice captured at a site in the scrub after Hurricane Opal in 1995 had been previously captured in the dunes before the storm (Swilling *et al.* 1998), suggesting that mice take refuge in the scrub. Since all of the mice inhabiting the dunes before the storm could not be marked, it is not known how many of the mice found in the scrub after the storm originated in the dunes; 47% is a very conservative minimum estimate.

It is a likely possibility that when there has been a long period of time without a hurricane, dunes function as a population source and scrub functions as a sink. After a major hurricane, however, dune habitat is destroyed and beach mouse persistence critically depends on the dune habitat recovering before the populations taking refuge in the scrub decline to extinction. If dune habitat does indeed recover before the mice in the scrub are gone then they can recolonize the dunes, and populations will persist in relative safety until the next hurricane occurs.

Increasing the rate of recovery of dune habitat is a seemingly effective method of conserving beach mice. Planting native vegetation and establishing sand fences (sand fences are structures that intercept drifting grains of sand, thereby hastening dune formation) on beaches affected by hurricanes are effective methods of accelerating the recovery of dune habitat. Dune restoration is practiced both inside and outside of the range of beach mice. It is somewhat ironic that beach mice benefit from a practice partially motivated by attracting tourism, which increases the demand for hotels and condominiums constructed upon their habitat. Nonetheless, it is not apparent that increasing the rate of dune recovery is more effective than decreasing the rate of population decline in the scrub. The latter could be accomplished through supplemental feeding or vegetation management. The most effective conservation strategy will depend on the relative contribution of source habitat recovery rate and the rate of population decline in the sink to population viability, as well as the relative cost of implementing each strategy. Another part of this management dilemma concerns the frequency of hurricanes. Again, it is not apparent whether the relative effectiveness of managing dunes, rather than scrub, changes as hurricane frequency changes. The effects of global climate change

on hurricane frequency and intensity is a very active area of research (Emanuel 2005; Landsea 2005; Trenberth 2005; Webster *et al.* 2005).

Riparian vegetation and floods

The banks of rivers and streams have unique substrate, moisture, nutrients, and light availability compared with the landscapes in which they are embedded. These physical conditions permit vegetative communities that are often quite distinct from their upland counterparts (Malanson 1993). Indeed, riparian zones are often afforded special legal protection from activities such as logging and farming that would otherwise alter plant composition. Such regulations are intended to conserve biodiversity and provide cleaner water.

We can readily imagine the source–sink dynamics of riparian plants. Species that are well adapted to the physical conditions adjacent to rivers and streams thrive; riparian zones function as sources for these plants. Propagules from such plants are dispersed into the uplands where they are ultimately out-competed by other species. Thus, upland areas function as sinks for riparian-adapted plant species.

This source–sink relationship can be severely disrupted by a flood. Plants in the “good” habitat along the river bank will be killed by a flood and the habitat structure (substrate, moisture, and nutrients) will be temporarily altered. Plants in the “poor” upland habitat will be relatively unaffected by the flood, and can be the source of immigrants to the recovering riparian habitat. When viewed over a time scale of a couple of hundred years, the presence of some riparian-adapted plants along a particular river may be credited to upland “sinks.”

Parks and fires

If unprotected habitats surrounding a park are sufficiently degraded, then such areas may function as population sinks. Habitats inside the protected area may normally function as population sources, but if fires have historically been suppressed, then potential fuel materials can accumulate, and the ensuing fires can be catastrophic. In this case, the putative source may be destroyed and long-term metapopulation persistence will be contingent upon recolonization from fire-resistant sinks outside the park into the recovering sources.

Model

Various model formulations could have been used to explore the interactions among (1) the rate of habitat recovery in the source, (2) the rate of population decline in the sink, and (3) the frequency of disturbances affecting the source. We chose to use population viability (i.e., the probability of persisting

over a given time period) as a response variable because of its (sensible) appeal to conservation biologists. Modeling population viability typically requires explicit consideration of demographic and/or environmental stochasticity because these often play a prominent role in extinctions. We also wanted a demographic model that tracks population size so that we could efficiently simulate immigration and emigration of individuals between populations. We further sought a model that would allow us to “manipulate” carrying capacity in order to simulate and evaluate the effectiveness of active (i.e., human-mediated) habitat recovery. Given these constraints, we elected to use a demographic sub-model described by Morris and Doak (2002: 127–133) but modified to suit our scenarios of interest. The demographic model simulates population size through time with both demographic and environmental stochasticity.

The mean growth rate over each time step is computed using a familiar logistic equation for the source population:

$$\mu_{\lambda}^{source} = e^{r \left(1 - \frac{N_t^{source}}{K_t} \right)} \quad (7.1)$$

where r is the intrinsic growth rate of the source population, which must be positive because births must exceed deaths if the habitat is a source. For all simulations, we fixed this value at 0.1. N_t is the current population size in the source and K_t is the current carrying capacity. With every time step, K increases by a given amount, hereafter denoted ΔK_{source} . Thus, for the sake of simplicity, the value of K is assumed to increase monotonically through time, and the rate of change is given by ΔK_{source} . The value of ΔK_{source} is systematically varied because the rate of habitat recovery in the source is one of three parameters being explored in this study. ΔK_{source} is varied from 25 to 250 individuals/year in increments of 25. There is no need to specify an upper limit to K because it is constrained by disturbances: K is reset to 500 individuals every time a disturbance occurs.

The sink population is not modeled with density dependence; rather, the population experiences exponential decline

$$\mu_{\lambda}^{sink} = e^r \quad (7.2)$$

when r , the growth rate of the sink, is negative. Since the population growth rate in the sink is another parameter under examination, it is also systematically varied. The growth rate in the sink varies from 0 to -0.1 in increments of 0.01.

Note that Eqs. (7.1) and (7.2) yield the mean growth rate for source and sink populations, respectively. Since we wish to simulate stochasticity (in order to generate a probability of persistence), it is necessary to also compute the variance around these means. This is done by computing both demographic and environmental contributions to the variance of mean growth rate using:

$$\sigma_{\lambda}^2 = \frac{1}{N_t} + 0.05. \quad (7.3)$$

Note that the first term on the right side of this equation contains population size in the denominator, indicating increased variance as population size decreases. This simulates demographic stochasticity. The second term on the right side of Eq. (7.3) is a density-independent constant that simulates environmental stochasticity.

With target values for the mean and variance of growth rates in both source and sink populations on hand, it is necessary to convert these quantities to an exponential scale because this is how random deviations subsequently enter the model. The mean and variance of a log-normal scale are related to the mean and variance of a normal variable being exponentiated by

$$\mu_r = \log(\mu_{\lambda}) - \frac{1}{2} \log\left(\frac{\sigma_{\lambda}^2}{\mu_{\lambda}^2} + 1\right) \quad (7.4)$$

$$\sigma_r = \sqrt{\log\left(\frac{\sigma_{\lambda}^2}{\mu_{\lambda}^2} + 1\right)}. \quad (7.5)$$

Projecting population size one time step forward is performed by

$$N_{t+1}^{source} = N_t^{source} e^{(\mu_r + (\sigma_r \varepsilon))} + (\alpha N_t^{sink}) - (\beta N_t^{source}) \quad (7.6)$$

$$N_{t+1}^{sink} = N_t^{sink} e^{(\mu_r + (\sigma_r \varepsilon))} - (\alpha N_t^{sink}) + (\beta N_t^{source}). \quad (7.7)$$

Since ε is a zero-mean, unit-variance random normal deviate in an exponent, the growth rates are log-normal (a desired outcome because this is frequently found in nature). μ_r is used in the exponent so that the mean growth rate matches μ_{λ} . Migration is simulated by moving a fixed proportion of individuals from the sink into the source (α), and from the source to the sink (β).

These procedures produce floating point variables for population size, which is biologically unrealistic (populations are composed of discrete individuals, not fractions thereof) and can result in a significant underestimation of the probability of extinction. Thus, a "floor" command is used at every iteration that eliminates all fractional individuals and restores population size to an integer value.

We simulated disturbances with a Poisson process. The mean return interval of the disturbance was systematically varied from 10 to 20 years in 1-year

increments. When a disturbance occurs, 80% of the source population is immediately lost and the carrying capacity of the source is set to 500 individuals (thereafter increasing according to ΔK_{source}). The sink population is unaffected by disturbances. Both populations were initiated with 200 individuals and simulated for 100 time steps (years). If no individuals remained at the end of the simulation, an extinction event was recorded. The probability of extinction was estimated by repeating each simulation 1,000 times.

We ran two sets of simulations with different source–sink migration rates in order to assess the robustness of our results. Our focal simulations were run with 6% of the sink population moving into the source (α , Eqs. 7.6 and 7.7), and 8% of the source population moving into the sink (β , Eqs. 7.6 and 7.7). These migration rates were increased to 10% and 12%, respectively, in the second set of simulations. We also performed separate simulations of systems both with and without an associated sink in order to provide a first approximation of the importance of sinks in the simulated environment. Unlike all other simulations that return the probabilities of persisting for 100 “years,” these populations were simulated until extinction occurred, thereby yielding persistence time.

Results

In our model, the presence of a sink population greatly enhances the persistence of the source population (Fig. 7.2). Indeed, median source population persistence time increases from 38 “years” ($SD = 14$, $N = 1,000$) when there is no associated sink (but individuals continue to emigrate from the source) to 135 “years” ($SD = 102$, $N = 1,000$) when a sink is present. These results are qualitatively consistent with previous findings that sinks enhance metapopulation persistence (Gyllenberg *et al.* 1996; Foppen *et al.* 2001). Indeed, since our putative source population cannot persist without the presence of the sink, these results agree with other findings that two sinks can sustain one another (Jansen and Yoshimura 1998). Results from the systematic variation of three model parameters are shown in Figure 7.3. These results were obtained with $\alpha = 6\%$ and $\beta = 8\%$. Not surprisingly, the probability of metapopulation extinction decreases as

1. the rate of habitat recovery in the sink increases
2. the population growth rate of the sink increases
3. the frequency of disturbances decreases.

For all levels of disturbance frequency, the effect of increasing the rate of population growth in the sink from -0.1 to 0 had a much greater effect on the probability of extinction than increasing the carrying capacity (rate of habitat

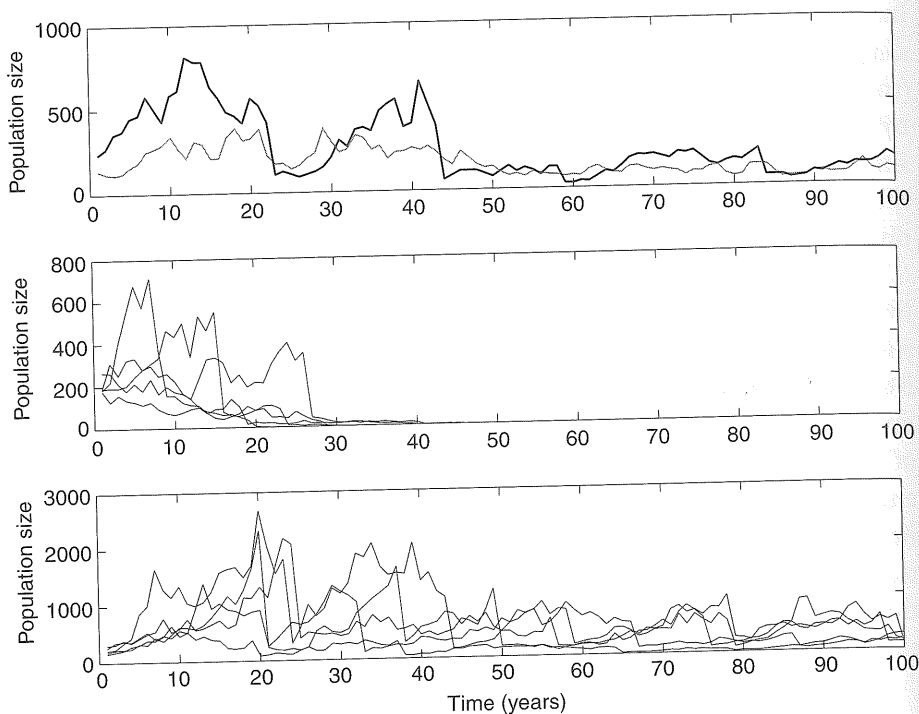


FIGURE 7.2.

Population trajectories when mean disturbance intervals is 20 years, habitat recovery rate in the source (ΔK_{source}) is 250, and population growth rate in the sink is -0.02 . All populations were initiated with 200 individuals. The top panel shows a single trajectory of the source population (heavy line) and the sink population (light line). The middle panel shows five replicate source populations when 8% of animals emigrate from the source, but the sink is not present (all emigrants die and immigration from a sink does not occur). The bottom panel shows five replicate source populations that are connected to a sink. Here, 8% of animals emigrate from source to sink and 6% of animals in sink emigrate to source. Note different scales on the y-axis for the three panels.

recovery) of the source from 25 to 250 individuals/year. Indeed, the contour lines in Figure 7.3 are almost horizontal, indicating that the effect of the recovery rate of the source is small compared with the effect of the sink population's growth rate. The contour lines in Figure 7.3 become increasingly horizontal when moving from the upper-left panel to the bottom-right panel. This indicates that as disturbances become more frequent, the relative importance of source habitat recovery rate declines.

Figure 7.4 displays two extinction isosurfaces that differ only in the given source-sink dispersal rates. The surfaces depict the combination of three parameter values that result in a 90% chance of extinction (an isosurface can be thought of as a sheet inserted into a stack of panels from Figure 7.3, with the sheet connecting equal probabilities of extinction). Isosurfaces for smaller

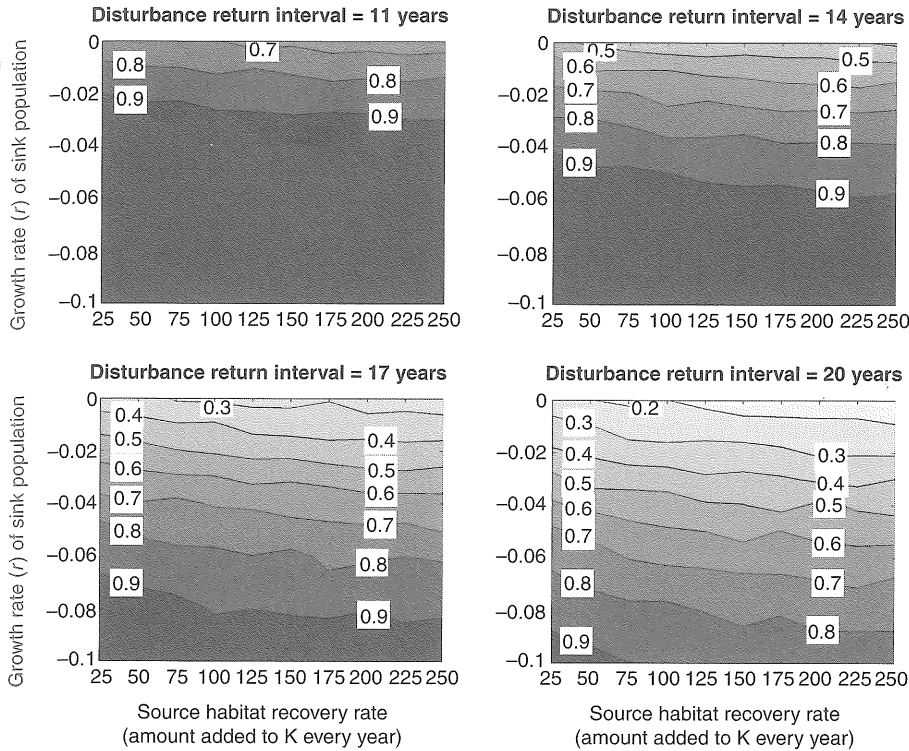


FIGURE 7.3.
Probability of extinction of a source-sink metapopulation.

extinction probabilities have similar shapes but occupy smaller slices of the parameter space, and are therefore not shown. Since the two surfaces differ only in the given dispersal rates between source and sink populations, their similar shape indicates that dispersal rates do not have a significant qualitative effect on the interactions between the parameters. Indeed, the position of the surfaces within the parameter space is very similar, indicating a small quantitative effect of dispersal rate.

Observing the bottom edge of a surface (i.e., a 10-year disturbance return interval) reveals that decreasing the rate of habitat recovery in the source can be “compensated” by a very minor increase in population growth rate in the sink in order to maintain a constant probability of extinction. The top edge of the surface suggests the same qualitative relationship. Quantitatively, however, there is a difference. The corner farthest from the viewer touches the edge of the box at approximately an 18-year disturbance return interval. As the recovery rate of the source decreases, a comparatively larger “compensation” in sink growth rate is needed at the 18-year mark. Stated simply, the surface is slightly twisted inside the box, revealing the slight three-way interaction between the factors. A very salient feature of Figure 7.4 is the approximately straight-line

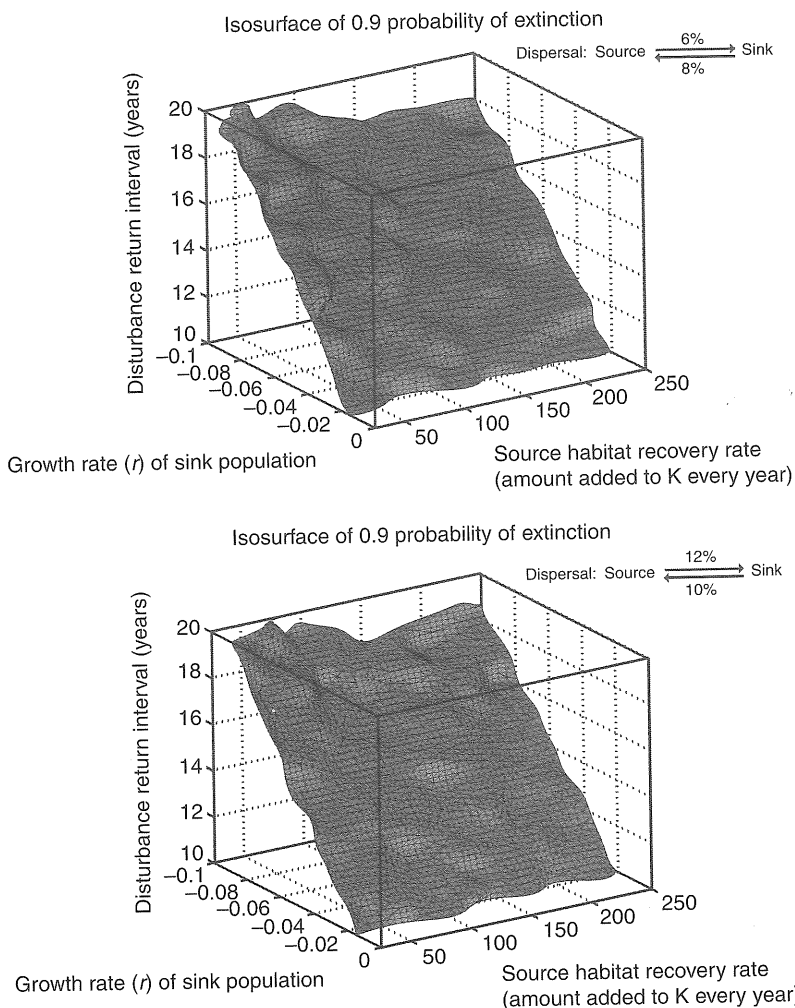


FIGURE 7.4.

The surfaces constitute combinations of three factors (model parameter values) that result in a 90% chance of extinction. Top and bottom panels differ in given dispersal rates between source and sink habitats.

relation between disturbance return interval and population growth rate in the sink. This is a significant interaction.

Discussion

The results presented here were obtained from a model of population dynamics that does not incorporate spatially correlated environmental stochasticity between source and sink populations. This is somewhat unrealistic in that both populations are within dispersal distance of one another and

are therefore likely to experience similar environmental phenomena such as droughts, seed masts, etc. Therefore, we ran other simulations with varying degrees of correlated environmental stochasticity to determine whether this created qualitatively different results from those reported here. It did not. Thus, for clarity and simplicity, we focus on output derived from a model with uncorrelated environmental stochasticity.

The rate of habitat recovery in the source (ΔK_{source}) and the rate of population decline in the sink (r_{sink}) have different units, so asking which has a greater impact on the probability of extinction is, to borrow a colloquialism, a bit like comparing apples and oranges. Transforming these rates to a common currency facilitates management decisions. One could then ask whether a given sum of money will "buy more persistence" if it is spent on improving the source habitat recovery rate versus increasing the population growth rate in the sink habitat. However, since the relative amounts of ΔK_{source} and r_{sink} that can be purchased with a given amount of money will vary from system to system (and possibly the amount of money to be spent within a particular system), we will not endeavor to transform ΔK_{source} and r_{sink} to a monetary scale. Instead, we limit our discussion to the effects of varying the model parameters, and petition managers to consider how much a given sum of money can change the rate of habitat recovery in a source versus how much the same amount of money can decrease the rate of population decline in the sink.

Our analysis demonstrates that increasing the rate of population growth in the sink has a much larger effect on population persistence than increasing the rate of post-disturbance habitat recovery in the source. This occurs over a range of values that managers are likely to encounter and potentially manipulate. Although a final decision to invest in source management versus sink management will depend on how many units of ΔK_{source} and r_{sink} can be purchased, sink management seems prudent, since even a modest improvement of r_{sink} has a larger effect on population viability than a big improvement in ΔK_{source} .

The precise frequency distribution of disturbances is often very difficult to estimate, especially when disturbances occur on a multidecadal time scale. To further complicate matters, global warming is suspected to augment some disturbance regimes. Our model shows that, as disturbance frequency increases, the relative effectiveness of increasing source habitat recovery rate decreases. Thus, management aimed at increasing the population growth rate of the sink will become even more prudent if disturbance frequency increases.

Conclusions

The values of biological definitions go far beyond their mere aesthetic appeal. Scientific progress is contingent upon our ability to communicate with

precision and robustness. Evolutionary biologists have been trying to define “species” for over a century, and unanimous acceptance of a definition does not seem forthcoming (Hey 2006). Ecologists face a similar predicament with “habitat.” The difficulty in arriving at an agreed-upon definition may have more to do with the progress of science itself than differences in the preferences of individual scientists. The target of a scientific definition is constantly moving as new discoveries add nuance to existing ideas.

Pulliam (1996) noted that the theory of island biogeography and metapopulation concepts helped legitimize the notion of “unoccupied habitat,” and that progress in source–sink dynamics is legitimizing notions that organisms can occupy “unsuitable habitat.” We hope our results will help legitimize the notion of “critical sink habitat.” Since proper management of imperiled species and associated legal conflicts can hinge on what is labeled “habitat,” we hope that the definition of habitat will include the types of sinks we described. Morris (2003) provides an excellent definition of habitat that can facilitate the conservation of species in complex landscapes with source–sink dynamics.¹

In this chapter, we described scenarios where sink populations rescue source populations, thereby saving a metapopulation from extinction. We demonstrated how improvement of sink habitat can be more effective in enhancing persistence than increasing the rate of recovery of source habitat. If society desires sustainable landscapes then we must be able to identify and protect sinks that rescue sources (see also Gimona *et al.*, Chapter 8, this volume). In some instances, this may not be achieved unless we adopt a definition of habitat that accommodates the critical role played by some sinks.

The source–sink dynamics described in this chapter occur over a time scale sufficiently long to include multiple disturbances. Thus, the consequences of degrading the quality of a sink that rescues a source may not become fully apparent for a very long time. The time lapse between sink degradation and metapopulation extinction constitutes a special form of the so-called “extinction debt” (Tilman *et al.* 1994), a term referring the time lag between habitat loss and fragmentation and ensuing extinction. Here, however, payment of the extinction debt occurs over a multiple-disturbance time period, which may be a substantially longer period than that caused by habitat loss and fragmentation. It is, therefore, conceivable that some recent extinctions and current metapopulation declines may be the result of events even more distant than commonly thought. As the first decade of the twenty-first century draws to a close with the release of this volume, it seems timely to note that it remains a matter of mere speculation to envisage the number of twenty-second century

¹ **Habitat:** A spatially bounded area, with a subset of physical and biotic conditions, within which the density of interacting individuals, and at least one of the parameters of population growth, is different than in adjacent areas (Morris 2003).

metapopulation extinctions that will ultimately be caused by events occurring in this decade. To end on a less macabre note, we offer hope in the form of a management directive: identify, protect and enhance sinks that rescue sources in dynamic environments.

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